The Bivariate Chen Distribution

The BvChen R Package

A toolkit for density, distribution, survival, dependence measures, EM-based estimation and goodness-of-fit analysis

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Abstract

This manual documents the BvChen package, an R implementation of the bivariate extension of the bathtub-shaped Chen distribution proposed by Gupta, Pundir, Sharma and Mesfioui (2022, *Life Cycle Reliability and Safety Engineering* 11:247–259), constructed from three independent latent Chen (2000) components through competing risks. The package provides the joint density, distribution and survival functions (including the singular component on the diagonal), random variate generation, marginal and conditional distributions, a Marshall–Olkin type survival copula with Kendall's τ, joint moments, maximum-likelihood estimation via the EM algorithm, Kolmogorov–Smirnov goodness-of-fit tests and Q–Q diagnostics. Complete univariate Chen building blocks are included. Every documented function is illustrated with executable examples whose live output appears directly beneath the code in this document.

Keywords: Chen distribution; bivariate lifetime model; Marshall–Olkin structure; competing risks; EM algorithm; survival copula; R.

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1 The univariate Chen distribution

The two-parameter Chen (2000) distribution has survival function

$$S(x) = \exp\{-\alpha(e^{x^\beta} - 1)\}, \qquad x > 0,\ \alpha, \beta > 0, \tag{1}$$

hazard rate $h(x) = \alpha\beta x^{\beta-1} e^{x^\beta}$ (bathtub-shaped or increasing) and quantile function $x_p = [\ln\{1 - \ln(1 - p)/\alpha\}]^{1/\beta}$.

2 The bivariate construction

Let X_0, X_1, X_2 be independent with $X_i \sim Ch(\alpha_i, \beta)$ and set $Z_1 = \min(X_0, X_1)$, $Z_2 = \min(X_0, X_2)$. Then $Z_1 \sim Ch(\alpha_1 + \alpha_3, \beta)$ and $Z_2 \sim Ch(\alpha_2 + \alpha_3, \beta)$ marginally. The joint law is a mixture: an absolutely continuous part on $\{z_1 < z_2\}$ and $\{z_1 > z_2\}$ carrying total mass $(\alpha_1 + \alpha_2)/(\alpha_1 + \alpha_2 + \alpha_3)$, and a singular component on the diagonal $z_1 = z_2$ of mass $\alpha_3/(\alpha_1 + \alpha_2 + \alpha_3)$.

2.1 Joint survival

With $t(z) = e^{z^\beta} - 1$,

$$S(z_1, z_2) = \exp\{-\alpha_1 t(z_1) - \alpha_2 t(z_2) - \alpha_3 \max(t(z_1), t(z_2))\}. \tag{2}$$

2.2 Survival copula

In terms of marginal survival levels $u = S_1(z_1)$, $v = S_2(z_2)$,

$$\hat{C}(u, v) = \min\left\{ u^{\frac{\alpha_1}{\alpha_1 + \alpha_3}} v,\ u v^{\frac{\alpha_2}{\alpha_2 + \alpha_3}} \right\}, \tag{3}$$

a Marshall–Olkin type copula whose dependence increases with the relative size of the common-shock parameter α_3 and is free of β.

2.3 Estimation by the EM algorithm

On the exponential scale $t_i = e^{z_i^\beta} - 1$ the latent model reduces to a Marshall–Olkin competing-risks scheme. The package alternates a numerical M-step maximising the observed log-likelihood in $(\alpha_1, \alpha_2, \alpha_3)$ at fixed β with a one-dimensional update of β; observed ties $z_{1i} = z_{2i}$ identify the common shock exactly.

Function documentation

`dbvch`	*The Bivariate Chen Distribution*

Description

Density (`dbvch`), joint CDF (`pbvch`), joint survival function (`sbvch`) and random generation (`rbvch`) for the bivariate Chen distribution built from three independent latent Chen components: $Z_1 = \min(X_0, X_1)$, $Z_2 = \min(X_0, X_2)$ with $X_i \sim Ch(\alpha_i, \beta)$.

Usage

```
dbvch(
  x,
  y,
  alpha1,
  alpha2,
  alpha3,
  beta,
  log = FALSE,
  component = c("full", "ac", "sing")
)

sbvch(x, y, alpha1, alpha2, alpha3, beta)

pbvch(x, y, alpha1, alpha2, alpha3, beta)

rbvch(n, alpha1, alpha2, alpha3, beta)
```

Arguments

`x`, `y`	vectors of non-negative quantiles; recycled to common length.
`alpha1`, `alpha2`, `alpha3`	positive shape parameters of the three latent Chen variables.
`beta`	positive common power parameter.
`log`	logical; if TRUE the log-density of the continuous part is returned (`-Inf` on the diagonal).
`component`	character selector for `dbvch`: `"full"` (default) evaluates the singular density on the diagonal and the continuous density elsewhere; `"ac"` evaluates only the absolutely continuous part (zero on the diagonal); `"sing"` evaluates only the diagonal singular density.
`n`	number of observations for `rbvch`.

Details

The joint law is a mixture of an absolutely continuous part living on $\{z_1 < z_2\}$ and $\{z_1 > z_2\}$, carrying total mass $(\alpha_1+\alpha_2)/(\alpha_1+\alpha_2+\alpha_3)$, and a singular part concentrated on the diagonal $z_1 = z_2$ with mass $\alpha_3/(\alpha_1+\alpha_2+\alpha_3)$. On $z_1 < z_2$ the density factorises as $f_{Ch(\alpha_1,\beta)}(z_1)\, f_{Ch(\alpha_2+\alpha_3,\beta)}(z_2)$.

Value

`dbvch`	numeric vector of density values.
`pbvch`	numeric vector, $P(Z_1 \le x, Z_2 \le y)$.
`sbvch`	numeric vector, $P(Z_1 > x, Z_2 > y)$.
`rbvch`	an $n \times 2$ numeric matrix with columns `z1`, `z2`, class `"bvch"`.

References

Chen, Z. (2000). A new two-parameter lifetime distribution with bathtub shape or increasing failure rate function. *Statistics & Probability Letters* 49, 155-161. Meintanis, S. G. (2007). Test of fit for Marshall-Olkin distributions with applications. *J. Statist. Comput. Simul.* 77, 171-179.

Examples (live output):

```
>
>
> # density pieces
> dbvch(0.5, 1.5, 1, 1, 1, 1.5) # region x < y
[1] 0.0005931645
> dbvch(1.5, 0.5, 1, 1, 1, 1.5) # region x > y
[1] 0.0005931645
> dbvch(1.0, 1.0, 1, 1, 1, 1.5) # singular diagonal density g(1)
[1] 0.02353232
> dbvch(1.0, 1.0, 1, 1, 1, 1.5, component = "ac") # 0 on diagonal
[1] 0
>
> # CDF from survival via inclusion-exclusion
> s <- sbvch(1, 2, 1, 1, 1, 1.5)
> s1 <- schen(1, 2, 1.5); s2 <- schen(2, 2, 1.5)
> all.equal(pbvch(1, 2, 1, 1, 1, 1.5), 1 - s1 - s2 + s)
[1] TRUE
>
> set.seed(1)
> sim <- rbvch(500, 1, 1, 1, 1.5)
> head(sim)
             z1 z2
[1,] 0.10617519 0.6814787
[2,] 0.08499553 0.1416199
[3,] 0.26448766 0.2644877
[4,] 0.25773862 0.2577386
[5,] 0.47240406 0.3201278
[6,] 1.22531700 0.2833057
> mean(sim[, "z1"] == sim[, "z2"]) # about a3/(a1+a2+a3) = 1/3
[1] 0.338
>
>
>
```

`dchen` *The Univariate Chen Distribution*

Description

Density, distribution function, quantile function, random generation, hazard function and maximum-likelihood fitting for the Chen (2000) distribution with shape parameter `alpha` and power parameter `beta`.

Usage

```
dchen(x, alpha, beta, log = FALSE)

pchen(q, alpha, beta)

schen(q, alpha, beta)

qchen(p, alpha, beta)

rchen(n, alpha, beta)

hchen(x, alpha, beta)
```

Arguments

`x, q`	vector of quantiles (non-negative).
`alpha`	shape parameter ($\alpha > 0$).
`beta`	power parameter ($\beta > 0$).
`log`	logical; if TRUE, the log-density is returned.
`p`	vector of probabilities.
`n`	number of observations.

Value

`dchen` gives the density, `pchen` the CDF, `schen` the survival function, `qchen` the quantile function, `rchen` generates random deviates, `hchen` gives the hazard rate and `fitchen` returns a fitted list with elements `alpha`, `beta` and `loglik`.

References

Chen, Z. (2000). A new two-parameter lifetime distribution with bathtub shape or increasing failure rate function. *Statistics & Probability Letters* 49, 155–161.

Examples (live output):

```
>
>
> dchen(1, alpha = 2, beta = 1.5)
[1] 0.2623826
> pchen(c(0.5, 1, 2), 2, 1.5)
[1] 0.5718313 0.9678249 1.0000000
> schen(1, 2, 1.5) # == 1 - pchen(1, 2, 1.5)
[1] 0.03217506
> qchen(c(0.25, 0.5, 0.75), 2, 1.5)
[1] 0.2623708 0.4457105 0.6521005
> hchen(seq(0.1, 2, length.out = 5), 2, 1.5)
[1] 0.9791627 3.5181490 9.0154268 24.3576403 71.7805109
> set.seed(1); x <- rchen(500, 2, 1.5)
> fit <- fitchen(x) # MLE close to (2, 1.5)
> fit$alpha; fit$beta
   alpha
2.219127
    beta
1.611988
>
>
>
```

`fitchen` *Univariate Maximum-Likelihood Fit of the Chen Distribution*

Description

Fits the two-parameter Chen distribution by numerical maximisation of the log-likelihood. Used internally to obtain starting values for `fitbvch`.

Usage

```
fitchen(x, start = NULL, tol = 1e-08, maxit = 100)
```

Arguments

`x`	non-negative numeric vector of observations.
`start`	optional named list with starting values `alpha`, `beta`; sensible defaults are chosen from the data.
`tol`	convergence tolerance used by `fitchen`.
`maxit`	maximum number of Newton iterations in `fitchen`.

Value

A list with components `alpha`, `beta`, `loglik`, `converged` and `se` (approximate standard errors from the numerical Hessian).

Examples (live output):

```
>
>
> set.seed(42)
> x <- rchen(300, alpha = 1.5, beta = 2)
> fit <- fitchen(x)
> fit$alpha; fit$beta; fit$loglik
   alpha
1.315234
    beta
2.028452
[1] -24.91924
>
>
>
```

`Marginals` *Marginal Distributions of the Bivariate Chen Law*

Description

Density and CDF of the marginals $Z_1 \sim Ch(\alpha_1 + \alpha_3, \beta)$ and $Z_2 \sim Ch(\alpha_2 + \alpha_3, \beta)$.

Usage

```
dmarg1(x, alpha1, alpha2, alpha3, beta)

dmarg2(y, alpha1, alpha2, alpha3, beta)

pmarg1(q, alpha1, alpha2, alpha3, beta)

pmarg2(q, alpha1, alpha2, alpha3, beta)
```

Arguments

`x, q`	non-negative quantiles.
`alpha1, alpha2, alpha3`	positive shape parameters of the three latent Chen variables.
`beta`	positive common power parameter.
`y`	quantile at which the second marginal is evaluated.

Value

Numeric vectors of density (`d*`) or probability (`p*`) values.

Examples (live output):

```
>
>
> dmarg1(1, 1, 1, 1, 1.5)
[1] 0.2623826
> dchen(1, 2, 1.5) # identical by construction
[1] 0.2623826
> pmarg2(c(0.5, 1, 2), 1, 1, 1, 1.5)
[1] 0.5718313 0.9678249 1.0000000
> pchen(c(0.5, 1, 2), 2, 1.5)
[1] 0.5718313 0.9678249 1.0000000
>
>
>
```

`Conditional` *Conditional Distributions of the Bivariate Chen Law*

Description

Conditional density (`dcond`) and conditional hazard (`hcond`) of Z_1 given $Z_2 = y$ (suffix 1) and of Z_2 given $Z_1 = x$ (suffix 2).

Usage

```
dcond1(z1, y, alpha1, alpha2, alpha3, beta)

dcond2(z2, x, alpha1, alpha2, alpha3, beta)

hcond1(z1, y, alpha1, alpha2, alpha3, beta)

hcond2(z2, x, alpha1, alpha2, alpha3, beta)
```

Arguments

`z1`, `z2`	values at which the conditional density/hazard of the response variable is evaluated (any non-negative value).
`alpha1`, `alpha2`, `alpha3`	positive shape parameters of the three latent Chen variables.
`beta`	positive common power parameter.
`x`, `y`	conditioning values (`x` conditions on $Z_1 = x$, `y` on $Z_2 = y$); must be positive.

Details

Given $Z_2 = y$, the conditional law of Z_1 has an atom at $z_1 = y$ of size $\pi(y) = g(y)/f_{m2}(y)$, where g is the diagonal singular density. Its continuous part is piecewise:

- $z_1 < y$: $f_{Ch(\alpha_1,\beta)}(z_1)$;
- $z_1 > y$: $f_{Ch(\alpha_2,\beta)}(y)\, f_{Ch(\alpha_1+\alpha_3,\beta)}(z_1)/f_{m2}(y)$,

with f_{m2} the marginal density of Z_2.

The conditional hazard of the absolutely continuous part is $h(z \mid y) = f_{Z_1|Z_2}(z \mid y)/P(Z_1 > z \mid Z_2 = y)$; it is `NA` exactly at $z = y$, where only the point mass lives.

Value

Numeric vector with attribute `"atom"` giving the conditional probability mass at $z_i = y$; the returned density itself refers to the strictly-off-diagonal part. `hcond*` return the conditional hazard rate of the absolutely continuous part (`NA` exactly at $z = y$).

Examples (live output):

```
>
>
> # conditional density of Z1 given Z2 = 1, below and above y = 1
> round(dcond1(seq(0.4, 1.8, by = 0.2), 1, 1, 1, 1, 1.5), 4)
[1] 0.9162 1.0235 0.9648 0.7314 0.1471 0.0107 0.0002 0.0000
attr(,"atom")
[1] 0.08968704
> attr(dcond1(1, 1, 1, 1, 1, 1.5), "atom") # point mass at z1 = y
[1] 0.08968704
>
> round(hcond1(seq(0.4, 1.8, by = 0.2), 1, 1, 1, 1, 1.5), 4)
[1] 1.2218 1.8493 2.7441 NA 10.9045 16.5804 25.5936 40.1383
>
>
```

tau_bvch *Dependence Measures of the Bivariate Chen Law*

Description

Kendall's tau (Monte-Carlo estimate via $\tau = 4E[F(Z_1, Z_2)] - 1$) and a concordance comparison of two parameter triples. Dependence increases with the relative size of the common-shock parameter α_3 and does not depend on β.

Usage

```
tau_bvch(alpha1, alpha2, alpha3, beta, n = 50000, seed = NULL)

concordance_bvch(
  alpha1a,
  alpha2a,
  alpha3a,
  alpha1b,
  alpha2b,
  alpha3b,
  beta,
  n = 50000,
  seed = NULL
)
```

Arguments

`alpha1`, `alpha2`, `alpha3`
: positive shape parameters.

`beta`
: positive power parameter.

`n`
: number of Monte-Carlo draws for `tau_bvch`.

`seed`
: optional seed for reproducibility.

`alpha1a`, `alpha2a`, `alpha3a`
: first parameter triple of `concordance_bvch`.

`alpha1b`, `alpha2b`, `alpha3b`
: second parameter triple of `concordance_bvch`.

Value

`tau_bvch`: numeric estimate of Kendall's tau. `concordance_bvch`: TRUE if the first triple is concordance-smaller than the second.

`Moments` *Moments of the (Bivariate) Chen Distribution*

Description

Moments of the (Bivariate) Chen Distribution

Usage

```
mchen(k, alpha, beta)

mbvch(r, s, alpha1, alpha2, alpha3, beta, tol = 1e-09)

ebvch(alpha1, alpha2, alpha3, beta)

vbvch(alpha1, alpha2, alpha3, beta)

min_bvch(alpha1, alpha2, alpha3, beta)
```

Arguments

`k`	moment order for the univariate integral.
`alpha`	shape parameter ($\alpha > 0$) of the univariate Chen distribution.
`beta`	positive common power parameter.
`r, s`	non-negative orders of the moments $E[Z_1^r]$, $E[Z_2^s]$ and the cross moment $E[Z_1^r Z_2^s]$.
`alpha1, alpha2, alpha3`	positive shape parameters of the three latent Chen variables.
`tol`	stopping tolerance for the numerical integration.

Value

`mchen`	numeric, $E[Z^k]$ for $Z\ Ch(alpha, beta)$.
`mbvch`	numeric, cross moment $E[Z_1^r Z_2^s]$.
`ebvch`	named vector of means `c(EZ1, EZ2)`.
`vbvch`	2x2 variance-covariance matrix.
`min_bvch`	list describing the minimum $M = \min(Z_1, Z_2) \sim Ch(\alpha_1 + \alpha_2 + \alpha_3, \beta)$ with elements `alpha`, `beta`, and functions `d`, `p`, `q`.

Examples (live output):

```
>
>
> mchen(1, 2, 1.5) # E[Z], Ch(2, 1.5)
[1] 0.4707204
> mean(rchen(200000, 2, 1.5)) # agrees up to MC error
```

```
[1] 0.4701932
>
> mbvch(1, 1, 1, 1, 1, 1.5)
[1] 0.2498289
> ebvch(1, 1, 1, 1.5)
     EZ1 EZ2
0.4707204 0.4707204
> vbvch(1, 1, 1, 1.5)
          Z1 Z2
Z1 0.06964704 0.02825116
Z2 0.02825116 0.06964704
>
> mn <- min_bvch(1, 1, 1, 1.5)
> mn$p(1) # P(min <= 1)
[1] 0.9942286
> pchen(1, 3, 1.5) # identical
[1] 0.9942286
>
>
```

`Likelihood` *Log-Likelihood and Profile Likelihood of the Bivariate Chen Model*

Description

Observed-data log-likelihood and profile likelihood for the common power parameter β.

Usage

```
profile_beta(
  z1,
  z2,
  betas = NULL,
  start = c(alpha1 = 1, alpha2 = 1, alpha3 = 1)
)

loglik_bvch(z1, z2, alpha1, alpha2, alpha3, beta)
```

Arguments

`z1`, `z2`	numeric vectors of paired observations (`z1[i]`, `z2[i]`).
`betas`	candidate beta values for the profile grid; a sensible default grid over (0, 20] is used if omitted.
`start`	named numeric vector with starting values for the alpha search inside `profile_beta`.
`alpha1`, `alpha2`, `alpha3`	positive shape parameters of the three latent Chen variables.
`beta`	positive common power parameter.

Value

`loglik_bvch` returns the scalar log-likelihood; `profile_beta` returns a list with the evaluated grid (`beta`, `prof_loglik`) and the optimiser `beta_hat`.

Examples (live output):

```
>
>
> set.seed(7)
> sim <- rbvch(300, 1, 1, 1, 1.5)
> loglik_bvch(sim[, "z1"], sim[, "z2"], 1, 1, 1, 1.5)
[1] -205.452
>
> pb <- profile_beta(sim[, "z1"], sim[, "z2"])
> pb$beta_hat # close to 1.5
[1] 1.523276
> head(pb$grid)
        beta prof_loglik
```

```
1 0.05000000 -1508.2561
2 0.06417844 -1384.5584
3 0.08237745 -1261.8931
4 0.10573713 -1140.5872
5 0.13572088 -1021.0718
6 0.17420709 -903.9121
>
>
```

fitbvch	*EM Estimation of the Bivariate Chen Distribution*

Description

Fits the four-parameter bivariate Chen distribution $(\alpha_1, \alpha_2, \alpha_3, \beta)$ to paired data with the EM algorithm. On the exponential scale $t = e^{z^\beta} - 1$ the model is a Marshall-Olkin competing-risks scheme, so each iteration re-estimates the α's in closed form (expected failures over exposure) and updates β by one-dimensional numerical maximisation of the observed log-likelihood. An observed tie $z_{1i} = z_{2i}$ identifies the common-shock component exactly.

Usage

```
fitbvch(z1, z2, start = NULL, tol = 1e-08, maxit = 200L, verbose = FALSE)

## S3 method for class 'bvchfit'
coef(object, ...)

## S3 method for class 'bvchfit'
print(x, digits = max(3L, getOption("digits") - 3L), ...)
```

Arguments

`z1, z2`	paired non-negative observations.
`start`	optional named list with starting values `alpha1`, `alpha2`, `alpha3`, `beta`; if omitted, values derived from marginal univariate fits are used.
`tol`	relative tolerance on the log-likelihood increase.
`maxit`	maximum number of EM iterations.
`verbose`	logical; print the iteration history.
`object`	an object of class `"bvchfit"`.
`...`	further arguments (unused).
`x`	an object of class `"bvchfit"`.
`digits`	number of digits passed to `round`.

Value

An object of class `"bvchfit"`: list with elements

`coefficients`	named vector `(alpha1, alpha2, alpha3, beta)`
`loglik`	observed log-likelihood at the solution
`iterations`	number of EM iterations performed
`converged`	logical convergence flag
`tau`	Kendall's tau implied by the fitted parameters
`n`	sample size
`n_ties`	number of observed ties $z_{1i} = z_{2i}$
`history`	iteration-wise log-likelihood values

References

Meintanis, S. G. (2007). Test of fit for Marshall-Olkin distributions with applications. *J. Statist. Comput. Simul.* 77, 171-179.

Examples (live output):

```
>
>
> set.seed(2024)
> sim <- rbvch(400, 0.8, 1.2, 0.5, 1.5)
> fit <- fitbvch(sim[, "z1"], sim[, "z2"])
> round(fit$coefficients, 3) # near (0.8, 1.2, 0.5, 1.5)
alpha1 alpha2 alpha3 beta
 0.776 1.368 0.645 1.572
> fit$loglik
[1] -264.4934
> fit$tau # dependence level
[1] 0.2379021
>
> # invariant to starting values
> fit2 <- fitbvch(sim[, "z1"], sim[, "z2"],
              start = list(alpha1 = 2, alpha2 = 2,
                           alpha3 = 2, beta = 1))
> all.equal(coef(fit), coef(fit2), tolerance = 1e-4)
[1] "Mean relative difference: 0.0001562438"
>
>
>
```

`gofbvch` *Goodness-of-Fit Tests and Q-Q Plots*

Description

Kolmogorov-Smirnov goodness-of-fit tests for the two marginals and for the minimum $\min(Z_1, Z_2)$, following the testing strategy of Meintanis (2007), together with Q-Q plots for Z_1, Z_2 and the minimum.

Usage

```
gofbvch(z1, z2, alpha1, alpha2, alpha3, beta)

qqbvch(z1, z2, alpha1, alpha2, alpha3, beta, plot = TRUE)
```

Arguments

`z1`, `z2`	paired non-negative observations.
`alpha1`, `alpha2`, `alpha3`, `beta`	fitted parameter values.
`plot`	logical; if TRUE (default) the Q-Q plots are drawn.

Value

`gofbvch` returns a list with elements `ks_z1`, `ks_z2`, `ks_min`: each an object of class `"htest"` produced by `ks.test`, comparing the data against the corresponding theoretical CDF.

`qqbvch` invisibly returns the sorted data and theoretical quantiles used in the plots, as a list with elements `z1`, `z2`, `min`.

References

Meintanis, S. G. (2007). Test of fit for Marshall-Olkin distributions with applications. *J. Statist. Comput. Simul.* 77, 171-179.

`plots` *Plots for the Bivariate Chen Distribution*

Description

Contour (`plot`) and perspective (`persp`) views of the absolutely continuous part of the joint density, plus a scatter-plot method for simulated data of class `"bvch"`.

Usage

```
## S3 method for class 'bvch'
persp(
  x,
  alpha1,
  alpha2,
  alpha3,
  beta,
  from = 0.05,
  to = 2.5,
  n.grid = 50,
  theta = -30,
  phi = 25,
  ...
)
```

Arguments

`x`	for `plot.bvch`: an `n x 2` matrix of class `"bvch"` (as produced by `rbvch`) or an object of class `"bvchfit"`; for `persp.bvch` the same types.
`alpha1`, `alpha2`, `alpha3`, `beta`	distribution parameters (required when `x` is a data matrix).
`from`, `to`	range of the evaluation grid on both axes.
`n.grid`	grid size per axis.
`theta`, `phi`	viewing angles passed to `persp`.
`...`	further graphical parameters.

Value

`plot.bvch` invisibly returns the evaluated density grid; `persp.bvch` invisibly returns the transformation matrix used.

Examples (live output):

```
>
>
> set.seed(11)
```

```
> sim <- rbvch(300, 1, 1, 1, 1.5)
>
> # contour of the AC density with the simulated scatter
> plot(sim, 1, 1, 1, 1.5, to = 2.5)
>
> # 3-D perspective of the absolutely continuous part
> persp(sim, 1, 1, 1, 1.5, to = 2)
>
>
```

References

Ahmed EA (2014) Bayesian estimation based on progressive type-II censoring from two-parameter bathtub-shaped lifetime model: an Markov chain Monte Carlo approach. J Appl Stat 41(4):752–768

Bhatti FA, Hamedani GG, Najibi SM, Ahmad M (2021) On the extended Chen distribution: development, properties, characterizations and applications. Ann Data Sci 8(1):159–180

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